VANDERBILT
School *of* Engineering

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Professor Mitchell
Introduction to Mechanical Engineering Design
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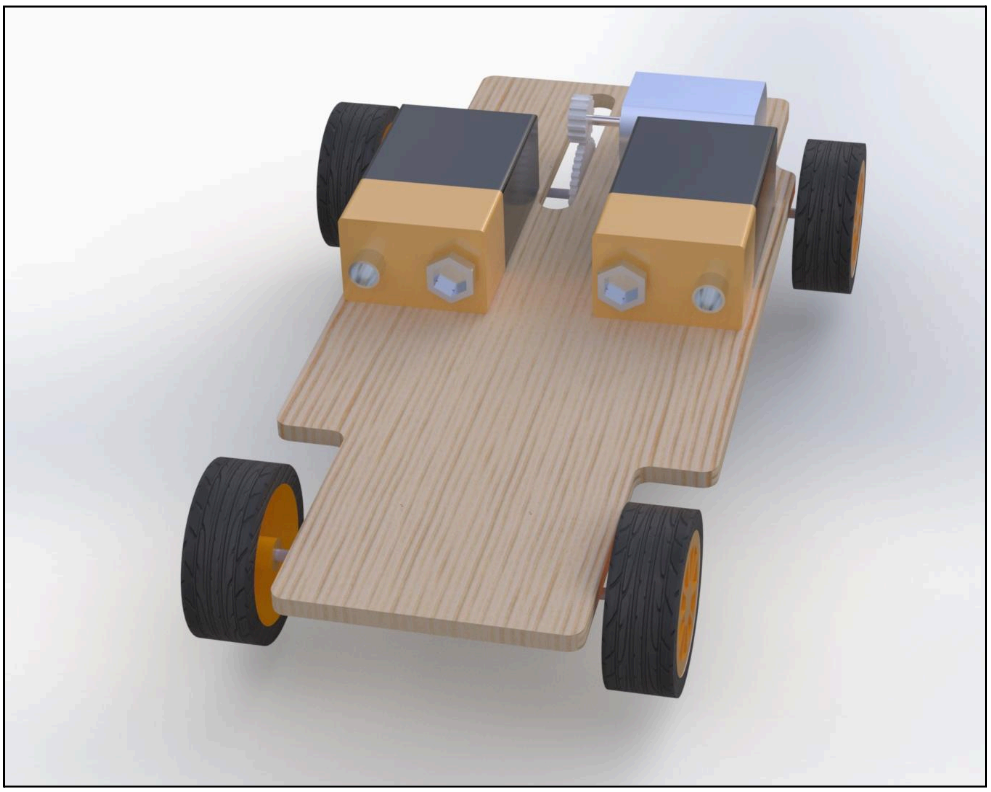
ME2160 Final Project – Electrically Powered Vehicle



1. Abstract

The team was tasked with designing an electrically powered vehicle that travels a distance of at least ten feet, using only parts from a kit of motors, gears, belts, and other mechanical components, as well as plywood from the Vanderbilt University Mechanical Makerspace. The team initially sought to construct a vehicle that could compete in the tractor pull category – a competition to pull the most weight – but upon reaching some challenges in the design process, opted to transition to a vehicle that travelled the distance in the shortest amount of time possible. After some iterations and testing, the team constructed a vehicle that utilized two batteries in tandem with a motor and gear system. The vehicle completed the distance in a time of 0.60 seconds and won the category for fastest vehicle. The differentiating factors in the final vehicle's design were a low gear ratio and two batteries that were wired together in series, amplifying the input voltage to the motor.

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2. Introduction

The goal of this project is to design an electrically powered vehicle that travels a distance of ten feet under its own power. The only allowed materials for the final design were parts from a gear, motor, and wheels kit from Amazon and ⅛" thick plywood from the Vanderbilt University Mechanical Makerspace. There were a variety of categories to design with in mind, such as the design that pulls the most weight, is the slowest, or occupies the least amount of volume. For this iteration of the project, the team designed the vehicle with the goal of creating the fastest car.

3. Background Research

In order to create the fastest car, there are a variety of design aspects to consider in order to optimize the speed of the car. We must examine topics from physics, mechanics, and circuits and incorporate them into the car's design.

One of the most basic aspects of the car's design is what dimensions and size the car should be. When designing for this, the simple physics equation $F = ma$, rearranged to yield $a = F / m$, tells us that we can increase the acceleration of the car by increasing the force that propels the car and decreasing the mass of the car. Therefore we will seek to use the smallest amount of material on the car and rig the motors to create the most amount of force – torque, in this case – to propel the car.

In order to transfer the power from the motor to the wheels, we must use a system of gears. To optimize the speed that the motor provides to the wheels, the speed ratio must be increased to its realistic maximum value. The speed ratio of a simple gear train can be calculated by: $speed\ ratio\ =\ \#\ of\ teeth\ on\ input\ gear\ /\ \#\ of\ teeth\ on\ output\ gear$. The problem specifications state that we must have a gear ratio of at least 3:1, meaning that the output gear must have at least three times the number of teeth as the input gear. Thus if we want to optimize the speed ratio, the output gear should have exactly three times the number of teeth as the input gear.

When designing the wiring for connecting the battery to the motor, we must only consider how the wiring should be if we were to use multiple motors. If we were to add multiple motors to the car, each would get half of the original voltage if wired in series and half of the original current if wired in parallel, which will lessen the overall power of the motor based on circuit rules. Therefore, to optimize power each motor should have its own dedicated battery.

A designer of this product completed a similar project to this during his junior year of high school, but the only allowed power supply was a mousetrap. During that project the designer learned that two key challenges were reducing friction between the axles and housing of the vehicle, and attempting to balance the weight of the car over the rear axle to maintain grip and prevent slipping. Similar design aspects will be incorporated into the car’s design.

4. Goal Statement

The goal of this project is to create an electrically powered vehicle that travels 10 feet in the shortest amount of time possible, utilizing only parts from the provided kit and plywood from the Vanderbilt Mechanical Makerspace.

5. Task Specifications

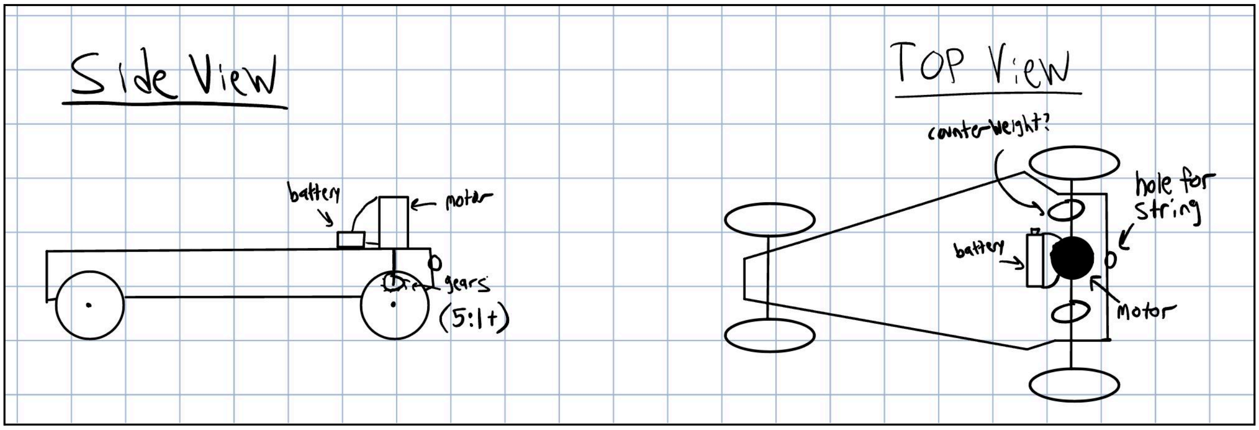
In order to achieve our goal of optimizing the speed of the vehicle, we must design with a specific set of specifications in mind:

- Minimize weight of the car by using as few materials as possible
- Optimize motor power by using the highest voltage battery without overloading the motor
- Maintain gear ratio as close to 3:1 as possible
- Reduce friction between axle and car housing
- Maintain rear wheel grip by shifting weight towards rear of vehicle
- Car must traverse at least ten feet

6. Design Description

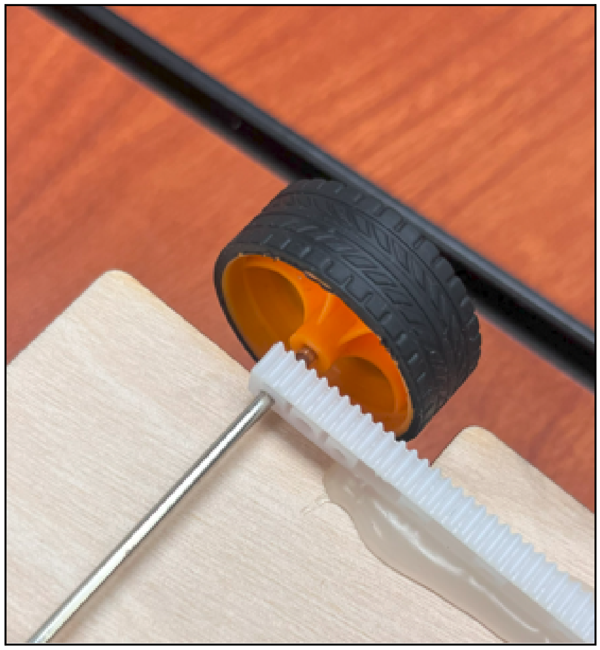
During the initial ideation phase for the vehicle, we believed that the best category we could design with in mind would be the tractor pull category, where we would attempt to pull the most weight by attaching a string to the rear of the vehicle. With this in mind, we would not attempt to optimize the speed of the car, but the torque of the car. This meant that we designed with a higher gear ratio in mind, as the motor would provide more torque to the axle but less speed. We would also mount the motor to the car's frame with the shaft perpendicular to the ground, so that

Figure 1: Initial Ideation of Tractor Pull Vehicle



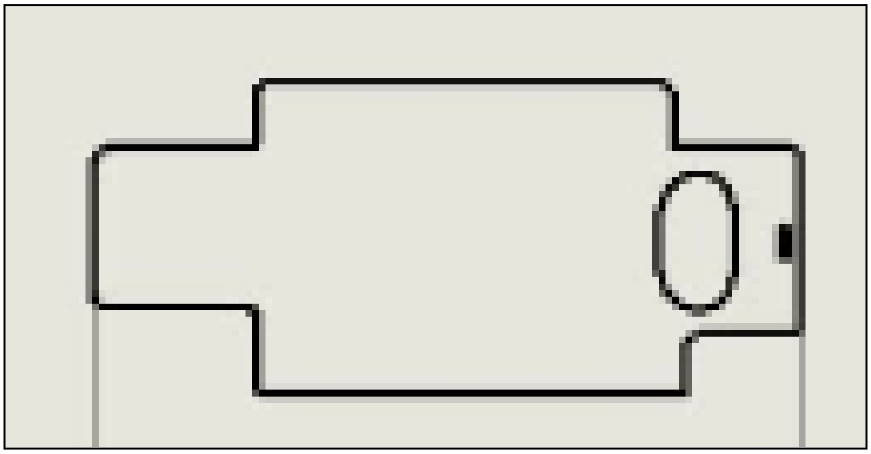
One of the first aspects of the vehicle's design we had to tackle was how to mount the frame of the vehicle to the axle, minimizing friction. As we combed through the kit to search for gears for the car's gear train, we uncovered a plastic rack that had multiple sets of holes bored into it on the ends of it. By chance, the diameter of the holes in the rack was slightly larger than the diameter of the axles provided in the kit. We observed that the metal axle and plastic rack had fairly low friction against each other, so we decided to use the racks to mount on the underside of the frame and pass the axles between the holes in the rack. This helped to reduce the amount of energy lost due to friction and ensures that the most amount of energy possible is transferred to the vehicle's wheels. Although the rack has one side with gears on it, the other side is flat and can be mounted to the underside of the frame using hot glue.

Figure 2: Photograph of Rack and Axle Design



Next came the design of the frame itself. As we wanted to lower the vehicle's center of mass and shift it as far towards the rear of the vehicle as possible, we then decided that the motor shaft would be parallel to the ground with a small spur gear mounted on the end of it, which would then be mated to a larger gear on the rear axle itself. Thus the initial frame design was created with the intent of occupying minimal area to reduce weight, as well as a slot for the gear to be exposed to the motor for. As required for the tractor pull competition, A smaller slot was also created towards the rear of the frame to allow for a string to be attached.

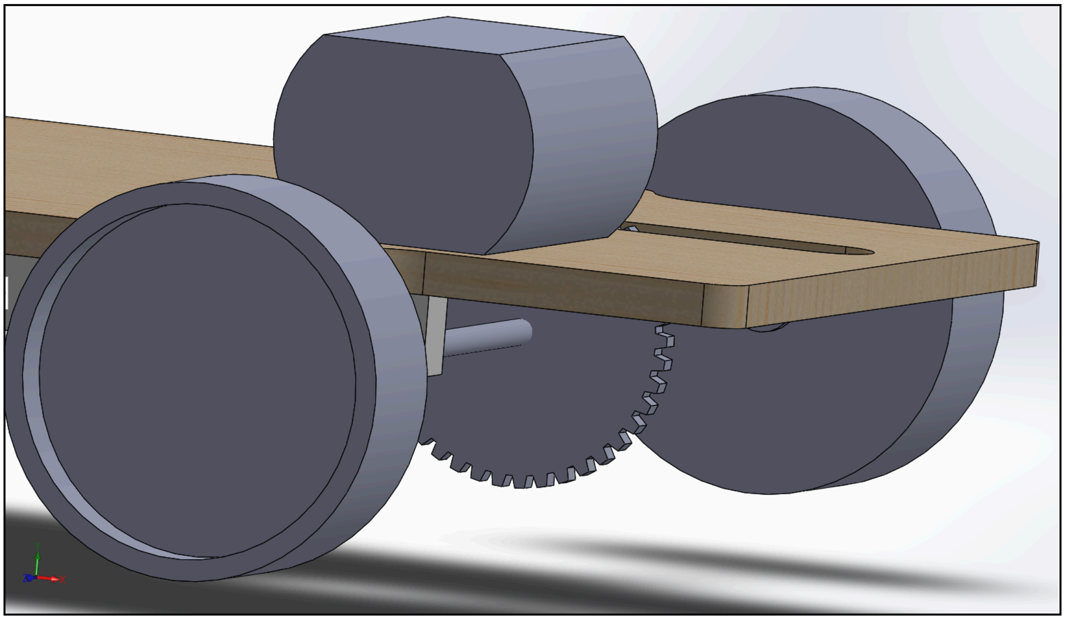
Figure 3: Initial Tractor Pull Car Frame Design



In order to optimize the car's performance and prevent the car from tipping due to the majority of mass being towards the rear of the vehicle, we realized that we needed to lower the car's center of mass as close to the ground as possible. Thus, the most massive parts of the car – the motor and battery – had to be mounted on the frame that was just above the car's axle. The car's axle was also very close to the ground in this configuration, due to the relatively small diameter of the wheels we were provided with. Once we selected a gear for the axle that matched our design criteria of a higher gear ratio – about five times as large as the small gear on the motor shaft, we ran into some issues.

The first issue was that the large diameter of the gear meant that the width of the slot that was created to expose the gear to the motor was too small for the gear to fit into. Any increase in the slot width would threaten the structural integrity of the frame, as the plywood would become very thin towards the edges of the frame. The large size of the gear also meant it had the potential to scrape against the ground as the car travelled, for the clearance above the ground is only the radius of the wheels.

Figure 4: Illustration of Gear Train Design

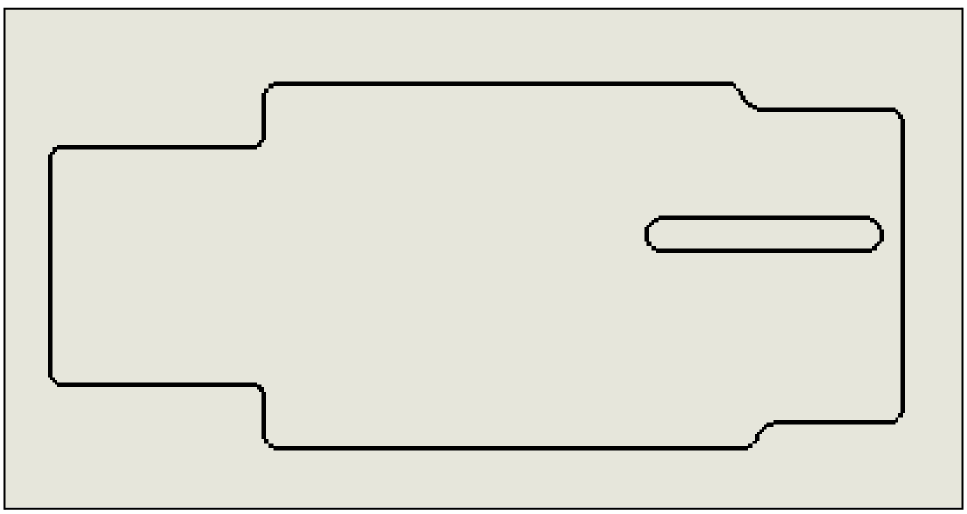


The diameter of the gear on the gear can only be as large as the diameter of the wheels, causing issues in selecting a large enough gear to achieve the desired gear ratio. Additionally, the space that was allocated for the motor to be mounted on would have to be shifted forward in order for the gears to mesh, threatening the grip of the rear wheels. Moreover, the large length of the slot allowed the gear on the axle to slide off the motor gear, leaving the car stationary as the motor failed to turn the axle's gear.

Thus we decided to alter our design category to alleviate these challenges – we shifted our focus to building the fastest car possible. Along those same lines, constructing the car that occupied the smallest possible volume could also be attempted. While some of the design aspects remained the same, such as reducing weight and optimizing the motor's power, we had to adjust our gear train to reflect these changes. Our background research concluded that we must use the smallest gear ratio possible to increase the vehicle's speed, and the guidelines for this project detail that the smallest gear ratio allowed is 3:1, so we decided to use that gear ratio to optimize speed.

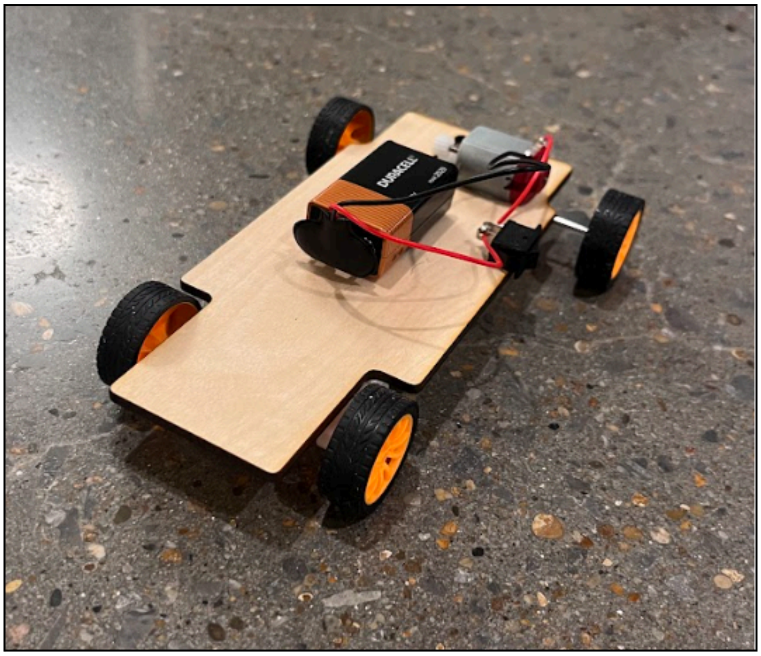
To solve the problem of the gear on the axle sliding around and becoming detached from the motor's gear, we rotated the slot 90 degrees from its previous configuration and decreased the width of the slot, forcing the gear to stay in place and be fully mated with the motor's gear. The slot for the string was also removed.

Figure 5: Revised Frame Design



This frame design successfully alleviated the issues faced in the previous iteration of the frame, and a more robust prototype was constructed. A 9-volt battery and switch were also mounted to the vehicle as a power supply, with the motor directly overhead of the rear axle gear. The wires between the motor, switch, and battery were soldered together to ensure a smooth connection.

Figure 6: Photograph of Proof of Concept Prototype

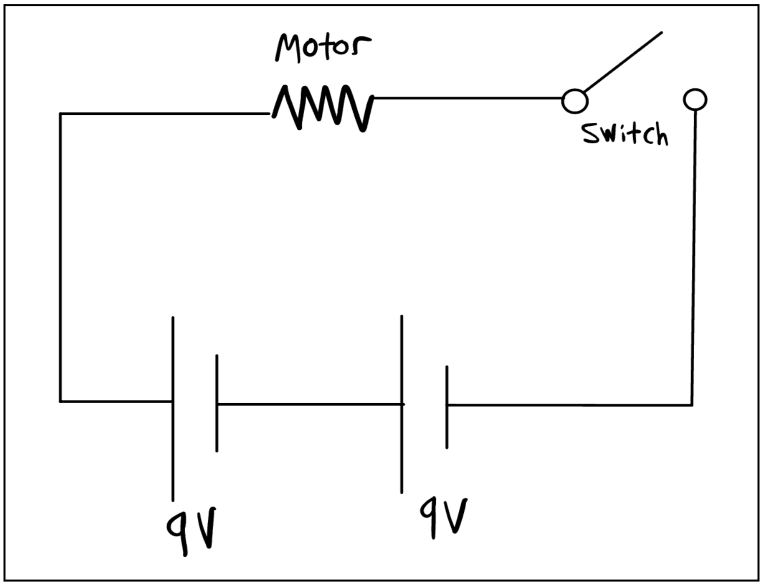


This prototype performed well on its first trial, completing the 10 yard course in about three seconds. However, there were still some aspects of the car that could be improved.

While the slot design did keep the rear axle's gear in place better than the previous design, it was still not perfect and required a manual reset before the car was set to run. Thus a small spacer was added to the side of the frame opposite to the motor to counteract this.

We then came to discuss how the car's speed could be increased. We thought about mounting another motor on the front of the car which would allow for four-wheel drive, also helping with grip, but decided that given the size of the two batteries needed, we would not have enough space on the frame to assemble that. However, along similar lines, we realized that we could increase the power that the motor receives by wiring another battery in series with the current one, increasing the power the motor receives and therefore also increasing the speed of the car. This would make the input voltage 18 V instead of the original 9 V.

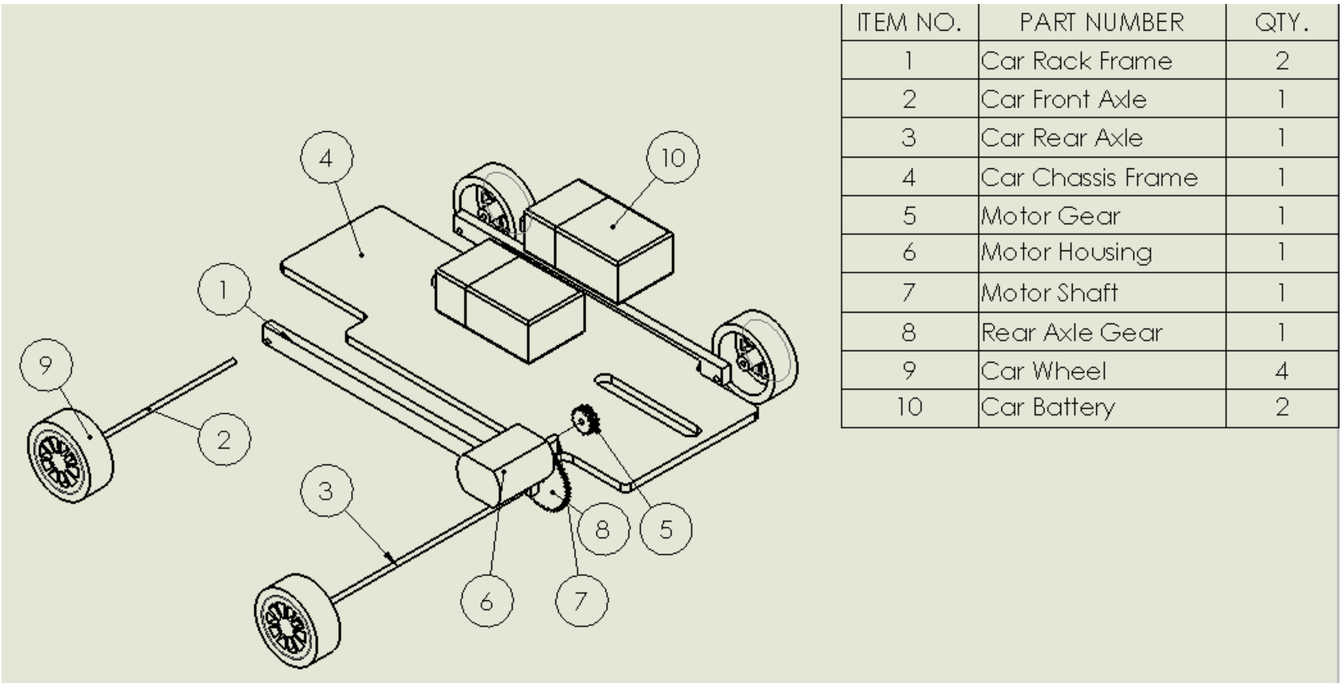
Figure 7: Final Vehicle Circuit Diagram



A second battery was then added to the vehicle's frame, and each battery was placed opposite of each other on the frame to ensure even weight distribution. The new circuit configuration successfully boosted the power of the motor and the speed of the car, causing the car to traverse the ten feet in significantly less time than the original proof of concept prototype.

7. Results

Figure 8: Final Vehicle Exploded Assembly Drawing



ITEM NO.	PART NUMBER	QTY.
1	Car Rack Frame	2
2	Car Front Axle	1
3	Car Rear Axle	1
4	Car Chassis Frame	1
5	Motor Gear	1
6	Motor Housing	1
7	Motor Shaft	1
8	Rear Axle Gear	1
9	Car Wheel	4
10	Car Battery	2

Figure 9: Photorealistic Rendering of Final Vehicle

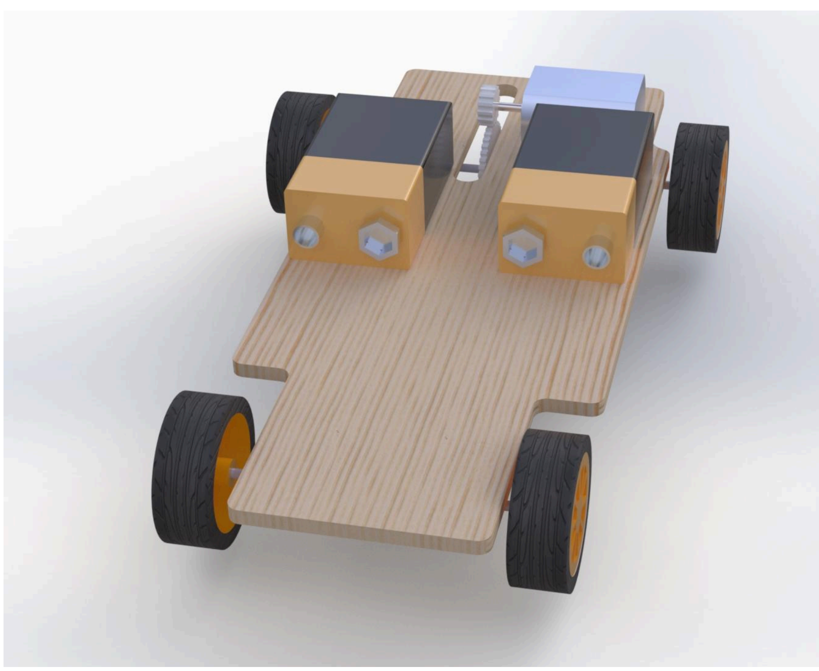


Figure 10: Final Vehicle Assembly Drawing

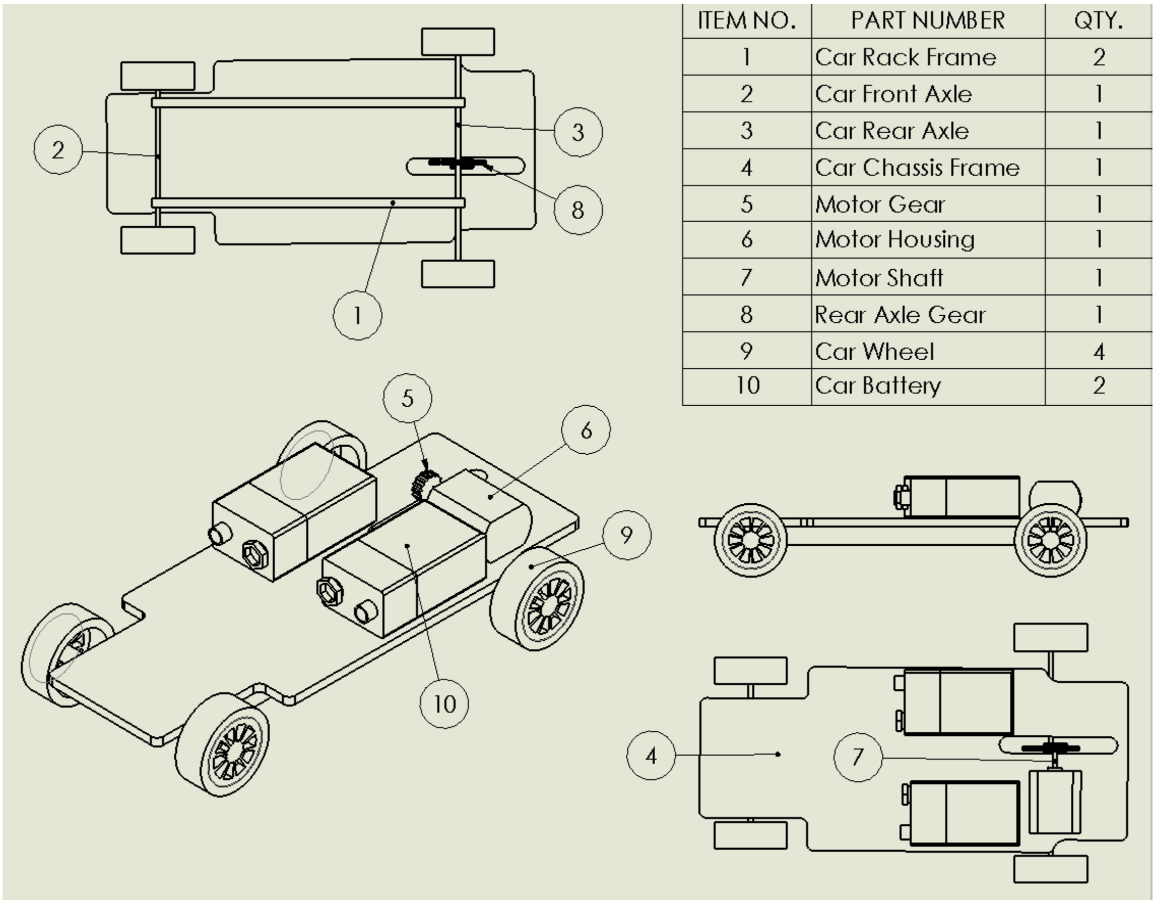


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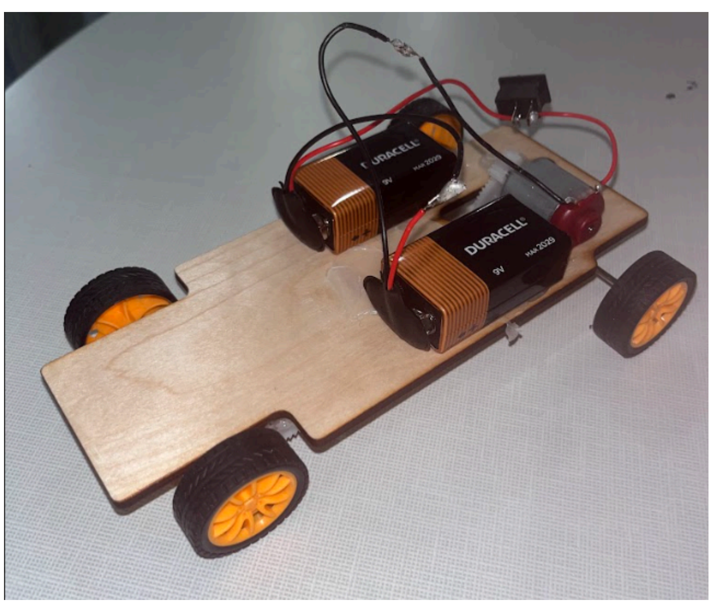
Figure 11: Photograph of Final Vehicle Design



Our vehicle proved successful in traversing the 10-foot distance, completing the course in 0.60 seconds, the shortest amount of time out of any vehicle in the test. The most significant aspect of our design is the wiring of two batteries in series, increasing the output voltage to the motor from 9V to 18V. This substantially increased the torque the motor provided to the wheels, increasing the speed of the vehicle. Despite the motor being rated for only 3-12V, the motor seems capable of operating at higher voltages for short amounts of time. Perhaps a third battery could be added to the circuit in an improved iteration of this design.

The car's gear system provides an effective transfer of power from the motor to the axle, as the gears always remain in sync in this configuration and there is no risk of slipping. Other car designs utilized an elastic band around two pulleys to propel the car, but the elasticity of the band seemed ineffective in transferring power.

Additionally, the wiring system could be constrained to the frame itself, preventing any circuiting issues from occurring. The switch is useful, but poses the risk of coming into contact with other metal items on the car and unintentionally opening the circuit. Placing an insulator around the switch could be an effective means of eliminating this issue.

8. Conclusion

We found this project an incredibly rewarding way to cap off our semester and bring something to reality. After creating many files in SolidWorks throughout the semester based on templates or guides, it was eye-opening to build a car from scratch and translate that into a CAD model from which we could create drawings and renderings. A hands-on project was also a welcome change of pace from the other CAD-intense sections of the course, as were the gear and pulley labs. We would highly recommend this class continues with this project.

As we reflect on the final state of our project, one of the biggest successes was shifting our design from focusing on the tractor pull to achieving the highest speed. We initially failed to recognize the constraints of the parts we were given from the onset of the project, finally coming to realize the tractor pull design would not work when we needed a gear with a bigger diameter than the wheels. Pivoting to designing with speed in mind allowed us to fine-tune our gear train to fit into a frame that brought together all the other components. Another success was recognizing the effectiveness of a series battery system. As we brainstormed how to increase the speed of the car after the proof of concept prototype, we initially believed we should add a second motor and battery to the front axle. However, we quickly recognized it would be difficult to find a gear that would sync well with the rear axle and run at the same speed. Thus we figured we could fit another battery onto the frame, and decided to try to wire it in series. This was the key to the final success of our project.

One of the most impactful parts of this project was getting familiar with the resources available to Vanderbilt students in the Wond'ry – laser cutting, soldering, hot glue, and wire cutters proved essential in the successful completion of this project. We now feel more familiar with all the resources in the Wond'ry, which will help us in completing future projects.

9. Appendix

Figure 12: Drawing of Battery

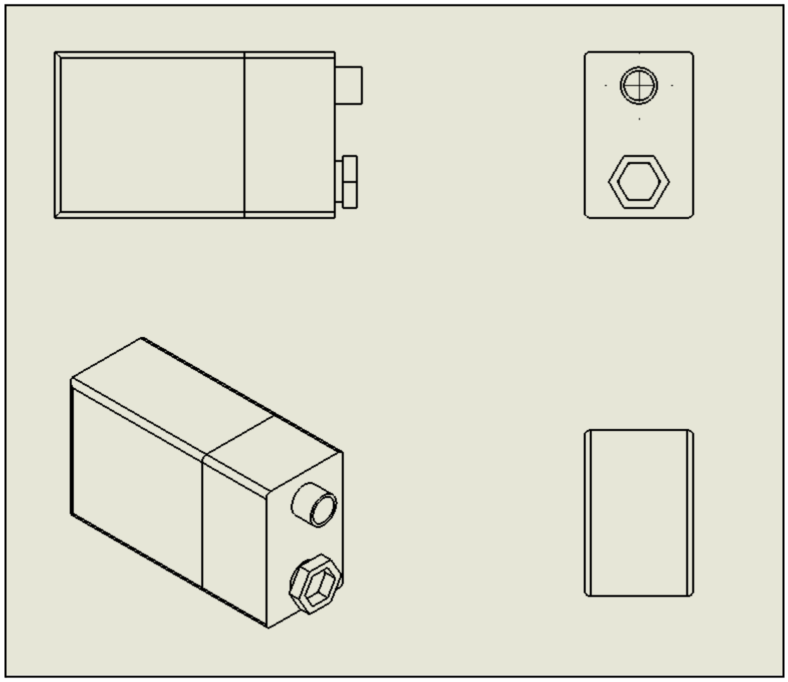


Figure 13: Drawing of Frame